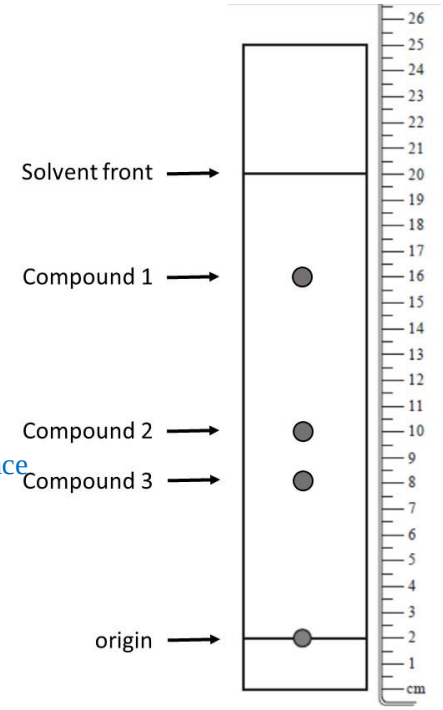
U3 SAC 1 Task 3 – Chapter 2-7 Test SOLUTIONS

SECTION A – Multiple choice

Use the following information to answer questions 1 and 2

A thin layer chromatography (TLC) plate was set up with a non-polar solvent (hexane) and a polar stationary phase (silica gel). A mixture containing several compounds was run, and the chromatogram below was obtained. A ruler was then placed next to the plate.



Question 1
The Rf value for compounds 1, 2 and 3 are:

A. 0.89, 0.56 and 0.44

B. 0.30, 0.40 and 0.70

C. 0.78, 0.44 and 0.33

D. 0.80, 0.50 and 0.40

The Rf value for the different compounds is calculated as Compound distance from origin ÷ Solvent front distance from origin. Therefore,

- Compound 1 = (16-2)÷(20-2) = 14÷18 = 0.78
- Compound 2 = (10-2)÷(20-2) = 8÷18 = 0.44
- Compound 3 = (8-2)÷(20-2) = 6÷18 = 0.33

Question 2
Based on the TLC results, which of the following statements is **true**?

A. Compound 1 and 2 are more polar than Compound 3

B. Compound 3 is the most polar compound

C. Compound 2 and 3 are less polar than Compound 1

D. Compound 2 is more polar than Compound 3

The given that the solvent is non-polar and the stationary phase is polar, the most non-polar (or least polar) compound, 1, will travel the furthest given its higher affinity for the mobile phase, while the least non-polar compound (or most polar), 3, will travel the least given its higher affinity for the stationary phase.

Question 3
The **equivalence** point for a titration of a weak acid with a strong base is most likely to be at a pH

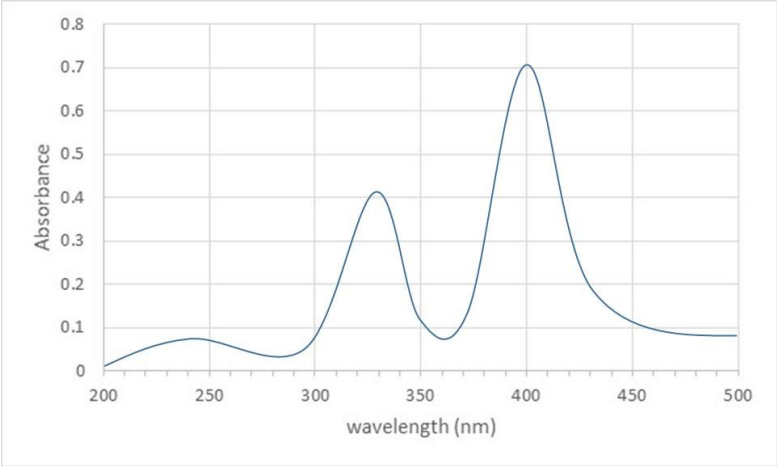
A. between 4 and 7

B. between 7 and 11

C. exactly 7

D. below 4

Question 4



The UV-visible spectrum of a solution of a certain compound is shown above. Consider the following statements about this compound and its UV-visible spectrum.

I. The spectrum is a result of electrons being excited from lower to higher energy levels.
II. The amount of light absorbed by a solution of this compound does not depend on the wavelength of light used.
III. The amount of light absorbed by a solution of this compound depends on its concentration.
IV. The concentration of a solution of this compound can only be determined by UV-visible spectroscopy at 400 nm.

Which of the above statements are **true**?

A. II and III

B. I and III

C. I, II and III

D. I, III and IV

Question 5

Analysis of the components of a mixture using gas-liquid chromatography normally involves

I. Measuring R_f values of the components
II. Measuring retention times of the components
III. Using a liquid as the mobile phase
IV. Using a gas as the mobile phase

Which of the above statements are **true**?

A. I and III

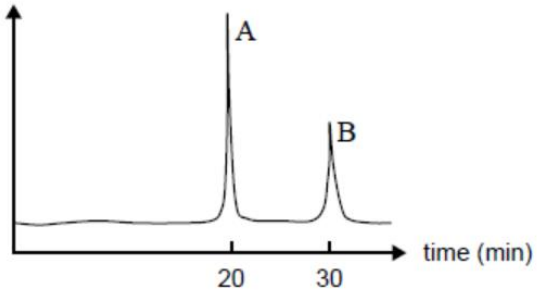
B. I and IV

C. II and III

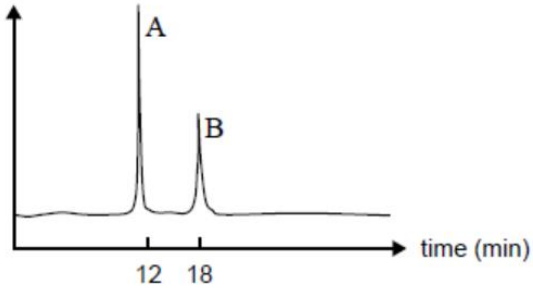
D. II and IV

Question 6

Chromatogram 1 was obtained by analysis of a sample of a mixture of two non-polar alkanes, A and B, using gas chromatography (GC). **Chromatogram 2** was obtained by analysing another sample of the same mixture, using the same GC instrument, under different experimental conditions.



chromatogram 1



chromatogram 2

Consider the following changes that could be made to the operating conditions for GC.

I. decreasing the pressure of the mobile phase
II. increasing the temperature
III. using a more polar column

Which of the changes would be most likely to produce **chromatogram 2**?

A. I only

B. II only

C. I and III

D. II and III

Question 7

An aqueous solution of nickel chloride is a clear green colour. The concentration of this solution is to be measured using UV-visible spectroscopy. An aqueous nickel chloride will

A. absorb mainly purple light and therefore allow purple light to pass through the solution.

B. absorb mainly purple light and therefore allow green light to pass through the solution.

C. absorb mainly green light and therefore allow purple light to pass through the solution.

D. absorb mainly green light and therefore allow green light to pass through the solution.

Question 8

An aqueous solution of copper sulfate appears blue in colour and its concentration can be determined by measuring its absorption at 600 nm. Which of the following errors will lead to a concentration **lower** than the real value:

I. Rinsing the colorimeter cell with water.
II. Preparing the calibration graph with a stock copper(II) sulfate solution that was 0.1 M less concentrated than what was written on the label.
III. Forgetting to take into account that the solution of unknown concentration was diluted before measuring its absorbance

A. I only

B. I and II

C. I and III

D. I, II and III

Question 9

Which of the following statements is **correct**?

A. The oxidation number for sulfur in $CaSO_4$ is +4 correct answer +6

B. The oxidation number for iron in Fe_2O_3 is +6 correct answer +3

C. The oxidation number for manganese in $KMnO_4$ is +1 correct answer +7

D. The oxidation number for sulfur in SO_3^{2-} is +4

Question 10

Which of the following shows a balanced equation for the reaction between sulfur dioxide and the dichromate ion in acidic solution.

A. $Cr_2O_7^{2-}$ (aq) + $3SO_2$ (aq) + $2H^+$ (aq) → $2Cr^{3+}$ (aq) + $3SO_4^{2-}$ (aq) + H_2O (l)

B. $Cr_2O_7^{2-}$ (aq) + $14H^+$ (aq) + $6e^-$ → $2Cr^{3+}$ (aq) + 7 H_2O (l)

C. SO_2 (aq) + 2 H_2O (l) → SO_4^{2-} (aq) + $2e^-$ + $4H^+$ (aq)

D. $Cr_2O_7^{2-}$ (aq) + SO_2 (aq) + $10H^+$ (aq) → $2Cr^{3+}$ (aq) + SO_4^{2-} (aq) + $5H_2O$ (l)

The half equation method must be used to balance this redox reaction. B shows the reduction half equation, C shows the oxidation half equation. These two half equations are combined to produce the overall balanced equation A. D is balanced but does not accurately represent this redox reaction.

Question 11
What is the molarity of a 40.0% solution of HCl which has a density of 1.20 g/mL?

A. 32.9 M

B. 13.2 M

C. 11.0 M

D. 3.28 M

If 1 mL of this solution has a mass of 1.20 g then 1000mL will have a mass of 1000 x 1.2 g = 1200 g
40.0% of this mass is taken up by HCl so the mass of HCl in 1L = 1200 g x 0.4 = 480 g
Convert mass to number of mole = m/M = 480 g / 36.5 g mol^{-1} = 13.15 mol L^{-1} = 13.2 M

Use the following information to answer Questions 12 and 13

In order to determine the salt (NaCl) concentration of potato chips, a 7.50 g sample of potato chips was blended with water and filtered. The solution was mixed with excess silver nitrate solution to precipitate the chloride as silver chloride. This was filtered washed and dried, and had a mass of 0.0276 g.

Question 12
The analytical technique described above is called

A. volumetric anlaysis

B. back titration

C. gravimetric analysis

D. percentage by mass

Question 13
The percentage by mass of NaCl in the potato chips was found to be

A. 0.606 % by mass

B. 0.0911 % by mass

C. 40.8 % by mass

D. 0.150 % by mass

n (AgCl) = 0.0276 g / 143.4 g mol^{-1} = 1.925×10^{-4} mol

n (NaCl) = n(AgCl) = 1.925×10^{-4} mol

m (NaCl) = 1.925×10^{-4} mol x 58.5 g mol^{-1} = 1.1126×10^{-2} g

% (NaCl) = (1.1126×10^{-2} g / 7.50 g) x 100 = 0.150% by mass

SECTION B – Short Answer

Question 1 (12 marks)

Iron ores are rocks from which metallic iron can be extracted. These rocks are usually found in the form of hematite (Fe_2O_3) or magnetite (Fe_3O_4). A quantitative analysis of iron in an unknown iron ore sample can be done by titration with a standard solution of potassium permanganate ($KMnO_4$).

a. Define quantitative chemical analysis and how it is different from qualitative chemical analysis.

Quantitative analysis is the determination of the **amount or concentration** of one or more components (ions, elements, or molecules) of a sample. (1)

Qualitative chemical analysis does not measure amounts but deals with the **identification of** components within a sample. (1)

2 marks

b. Potassium permanganate ($KMnO_4$) is a strong oxidizing agent that is inexpensive, readily available, and works as a chemical indicator. Potassium permanganate solutions are purple but in dilute solutions when the permanganate is **reduced** to Mn^{2+} the products result in a colourless solution. During a titration using a standardized solution of $KMnO_4$, when one drop excess of $KMnO_4$ is added, the solution will take on a pale purple colour.

i. Explain why, during a titration, a colourless solution does not change colour to purple until the $KMnO_4$ is in excess.

When the $KMnO_4$ is first added the reductants within the sample reduce the coloured MnO_4^- to colourless Mn^{2+}. The solution remains colourless (1)

When the $KMnO_4$ is in excess, there are no more reductants in the sample to reduce the purple MnO_4^- and so the solution appears purple. (1)

ii. Why is potassium permanganate not a primary standard?

A primary standard is a soluble solid compound **from which a solution of known concentration can be directly prepared**. It is very pure, with a consistent formula that does not change on exposure to the atmosphere, and has a relatively high molar mass. (1, understanding of what a primary standard is)

$KMnO_4$ is a strong oxidizing agent that will oxidise any traces of organic materials in water (it also decomposes in sunlight). **Its formula will change on exposure to the atmosphere and it's concentration can therefore not be accurately known.** (1 explanation for why $KMnO_4$ does not meet the requirements for being a primary standard)

iii. Briefly explain how a secondary standard solution of $KMnO_4$ would be standardized.

A solution of $KMnO_4$ would be prepared to an approximate concentration and then titrated against a standardized solution to determine its accurate concentration. (1)

2 + 2 + 1 = 5 marks

c. When the percentage of iron in a sample of iron ore is determined via titration it is important that all of the iron in the unknown sample is in the form of Fe^{2+}. To ensure that this happens, the sample is treated with tin(II) chloride, which **reduces** any small amount of Fe^{3+} that might be present to Fe^{2+}. In this process the Sn^{2+} is **oxidized** to Sn^{4+}. Any excess Sn^{2+} is then removed by reaction with mercury(II) chloride, which produces insoluble mercury(I) chloride.

i. Write a half-equation to represent the oxidation of Sn^{2+}.

Sn^{2+} (aq) → Sn^{4+} (aq) + $2e^-$
(1)– do not penalise if states are not included

ii. Write a net ionic equation to represent the reaction between Fe^{3+} and tin(II) chloride.
Fe^{3+} (aq) + e^- → Fe^{2+} (aq) (1) for reduction reaction

Sn^{2+} (aq) + 2 Fe^{3+} (aq) → Sn^{4+} (aq) + 2 Fe^{2+} (aq) (1) for balanced equation
(1) states

iii. Write a net ionic equation to represent the reaction between Sn^{2+} and mercury(II) chloride.

Sn^{2+} (aq) + 2 Hg^{2+} (aq) + 2 Cl^- (aq) → Sn^{4+} (aq) + 2 HgCl (s) (1) balanced equation

1 + 3 + 1 = 5 marks

Question 2 (8 marks)

The vitamin C concentration in a solution can be determined via redox titration using iodine solution and a starch indicator. As the iodine solution is added during the titration, the vitamin C (ascorbic acid, $C_6H_8O_6$) is oxidised to dehydroascorbic acid ($C_6H_6O_6$):

$$C_6H_8O_6 + I_2 \rightarrow C_6H_6O_6 + 2I^- + 2H^+$$

As long as vitamin C is present in the solution, the iodine is converted to the iodide ion very quickly. When the all the vitamin C is oxidized, excess iodine reacts with the starch indicator to form a blue-black complex. This is the endpoint of the titration.

In 2004 two 14 year old students, Anna Devathasan and Jenny Suo, decided to use this method to test the vitamin C concentration of their favourite fruit juices, including Ribena blackcurrant juice and Just Juice orange juice. They pipetted 20.0 mL aliquots of each juice sample into separate conical flasks. To each flask they added about 150 mL of deionised water and 1 mL of starch indicator solution. They then titrated this with a 5.00×10^{-3} M iodine solution. Their results are shown in Table 1 below:

Table 1: Titration results for investigation into vitamin C content of fruit juices.

Juice sample	**Average titre to reach end point**
Ribena blackcurrant juice	5.00 mL
Just Juice orange juice	16.36 mL

a. Calculate the concentration of vitamin C (ascorbic acid, $C_6H_8O_6$) in mg/100 mL, in the each of the juice samples.

Ribena Blackcurrant Juice

$n(I_2) = c \times v$
$= 5.00\times10^{-3}$ M $\times$ 5.00×10^{-3} L
$= 2.5\times10^{-5}$ mol (1) number of mole I_2

$n(C_6H_8O_6) = n(I_2) = 2.5\times10^{-5}$ mol (1) number of mole $C_6H_8O_6$ (mole ratio)

$m(C_6H_8O_6) = n \times M$
$= 2.5\times10^{-5}$ mol $\times$ 176 g mol^{-1}
$= 0.0044$ g (1) mass $C_6H_8O_6$

$c(C_6H_8O_6)$ = 4.4 mg in 20 mL
= 22 mg in 100 mL
= 22.0 mg/100 mL (1) mass $C_6H_8O_6$

Just Juice Orange Juice

Use $c_1v_1 = c_2v_2$

$c(C_6H_8O_6)$ in orange = (22.0 ÷ 5.00 mL) x 16.36 mL
= 72.0 mg/100 mL (1) $c(C_6H_8O_6)$ in orange juice

5 marks

b. Why did Anna and Jenny add 150 mL of deionized water to each flask?
Both the orange juice and the blackcurrant juice have a strong colour. The water is added to ensure that the endpoint (colour change) can be detected.

1 mark

c. The advertising slogan for Ribena blackcurrant juice stated that Ribena blackcurrant juice had four times the vitamin C of orange juice. If this information was found to be incorrect, the makers of Ribena blackcurrant juice could be fined (be made to pay) over $200,000 for providing false information to customers.

i. According to Anna and Jenny's results, should the makers of Ribena blackcurrant juice be fined? Discuss.
The company should receive a fine as the information in their advertisement was incorrect. Ribena contained less vitamin C than the orange juice.

ii. Describe one error that Anna and Jenny could have made that would cause **their result** for the concentration of vitamin C to be **lower than the actual value**.
The pipette may have been rinsed with water which would lower the concentration of vitamin C in the aliquot.

1 + 1 = 2 marks

FYI

Ribena-maker fined $217,500 for misleading vitamin C ads

5:00 PM Tuesday Mar 27, 2007

Auckland Region | Australasia | Business | ...

GlaxoSmithKline, the maker of Ribena, has been fined $217,500 after admitting it mislead customers about the vitamin C content of the blackcurrant drink.

The company appeared in Auckland District Court to face charges alleging 15 breaches of the Fair Trading Act.

It admitted that its cartoned Ready To Drink Ribena, which it claimed had 7mg of Vitamin C per 100ml, in fact had no detectable Vitamin C content.

The company also admitted it may have misled customers in advertisements saying the blackcurrants in Ribena syrup had four times the Vitamin C of oranges.

In court this afternoon, Judge Phil Gittos fined GlaxoSmithKline and ordered the company to place half-page corrective advertisements in the Herald, Dominion Post, the Press and Otago Daily Times.

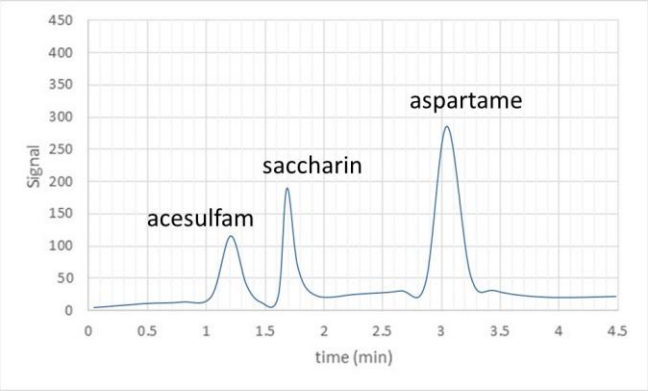
He said the advertisements' wording needed court approval and would have to run at least twice within the next month.

http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10431119

Question 3 (9 marks)

Aspartame, saccharin and acesulfam are artificial sweeteners used as a sugar substitute in food and beverages. They are sweeter than sugar, and used in many diet soft drinks. The concentration of these sweeteners in drinks can be determined using HPLC.

A 50µL solution containing aspartame, saccharin and acesulfam was injected into an HPLC column and the following chromatogram was obtained.

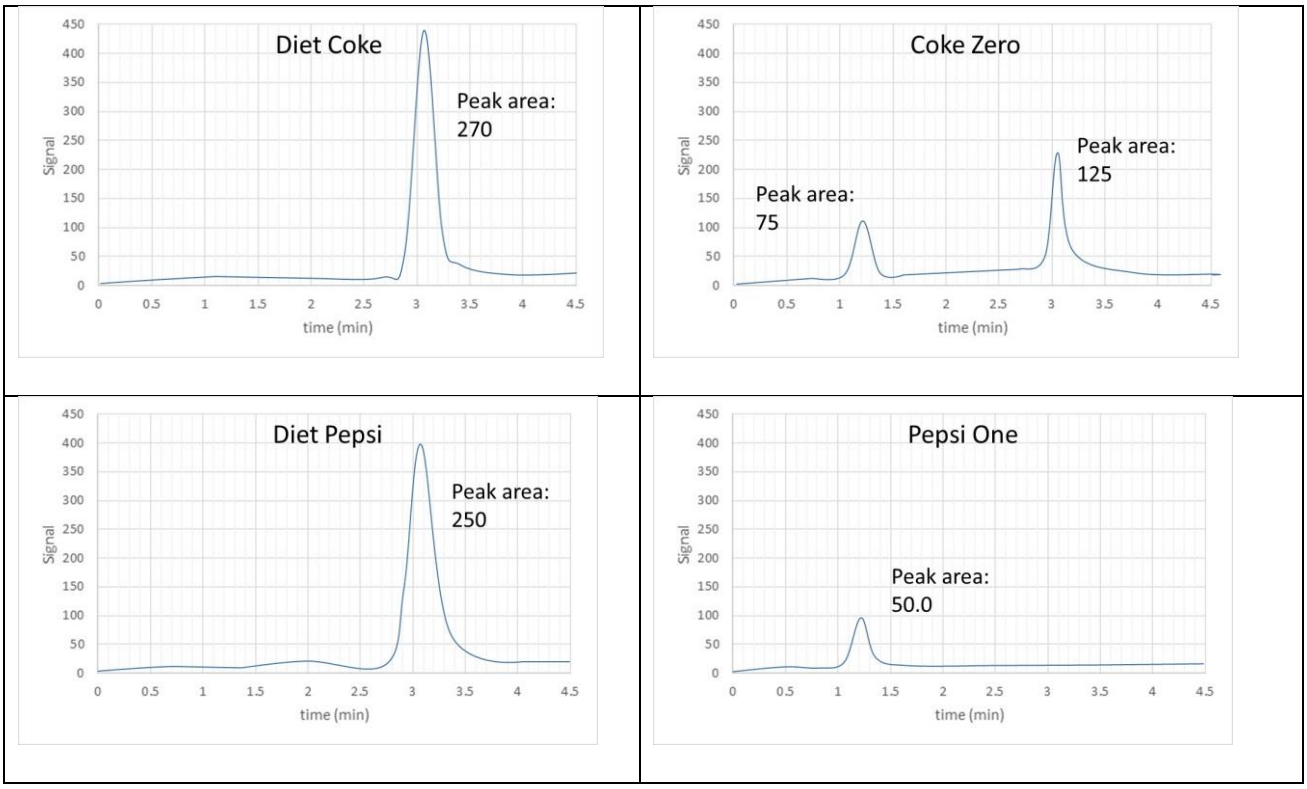


a. From these results, the retention time for the different sweeteners are:

Sweetener	**Retention time (min)**
Aspartame	3.05
Saccharin	1.7
Acesulfam	1.2

1 mark

b. To analyse the sweetener content of four different diet soft drinks, 50µL samples were injected into the HPLC column. Their chromatograms, with the corresponding area for each peak, are shown below



Based on these results, indicate with a tick the sweetener present in each soft drink

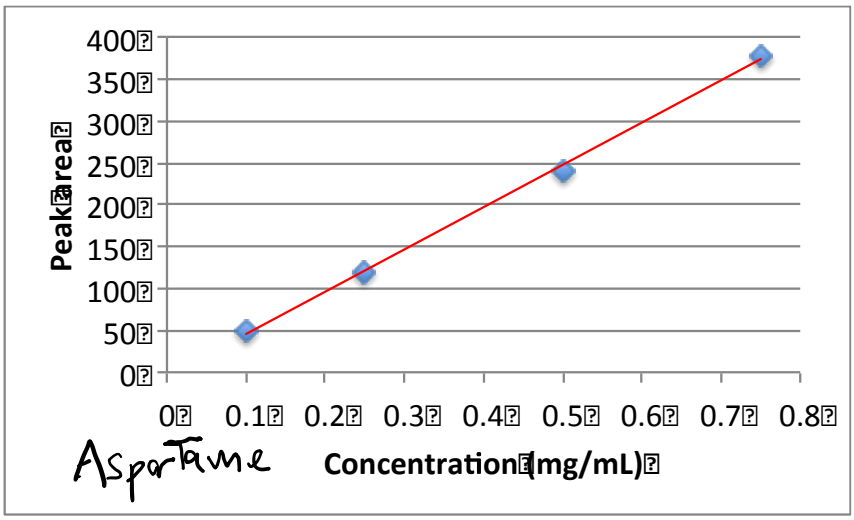
	Sweetener		
Soft Drink	**Aspartame**	**Saccharin**	**Acesulfam**
Diet Coke	✓		
Coke Zero	✓		✓
Diet Pepsi	✓		
Pepsi One			✓

4 marks

c. Five aspartame standard solutions containing 0.100 mg/mL, 0.250 mg/mL, 0.500 mg/mL and 0.750 mg/mL were prepared. 50µL of each sample was injected into the HPLC column. The chromatograms of the standard solutions each produced a single peak at a retention time of 3.05 minutes. The peak areas were measured and recorded in the table below.

Aspartame standard (mg/mL)	**Peak area**
0.100	50.0
0.250	120
0.500	240
0.750	380

i. Using the results for the 4 standard solutions, construct a calibration graph including a straight line of best fit. Clearly label the axes and use appropriate units.



ii. Using the calibration graph and the chromatograms for each of the soft drinks calculate the amount of aspartame present in each of them

Soft Drink	**Aspartame (mg/mL)**
Diet Coke	0.540 (range 0.530-0.550)
Coke Zero	0.250 (range 0.240-0.260)
Diet Pepsi	0.500 (range 0.490-0.510)
Pepsi One	0.000

iii. Taking into account that the acceptable daily intake of **aspartame** is 50.0 milligrams per kilogram (mg/kg), what is the maximum **volume** of Diet Coke that a person of 70.0 kg should drink per day?

Daily intake for a 70 kg person = 50.0 mg/kg × 70 kg = **3.50 g** of aspartame → 1 mark for this step

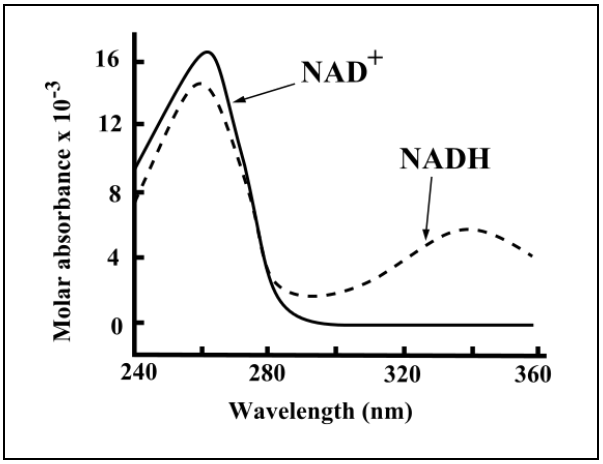
Aspartame in Diet Coke: 0.54 mg/mL

Therefore, the volume will be 3.50 g × 1×10^{-3} L ÷ 0.54×10^{-3} g = **6.48 L**. (range 6.36-6.60 L).

2 + 2 + 2 = 6 marks

Question 4 (8 marks)

Nicotinamide adenine dinucleotide (NAD) is a coenzyme found in all living cells. It exists in two forms, an oxidized and a reduced form abbreviated as NAD^+ and NADH, respectively. The concentration of NAD^+ and NADH can be determined using UV-visible spectroscopy. The individual spectra of NAD^+ and NADH are presented below.



a. Which wavelength could be used used for absorbance measurements to determine the concentration of:

i. A solution containing only NAD^+

260-270 nm

ii. A solution containing only NADH

Three possible correct answers: 260 nm;

340 nm

260 nm or 340 nm

1 + 1 = 2 marks

b. The concentration of NADH can be determined using the following equation:

$$[NADH] = Absorption_{340nm} \div 6620\ M^{-1}$$

The enzymatic reduction of NAD^+ to NADH was followed in solution by monitoring the absorption at 340nm. The values obtained are shown in the table below:

Time	0 min	1 min	2 min	3 min	4 min	5 min
$Absorption_{340nm}$	0.000	0.100	0.250	0.330	0.390	0.390

Taking into account that the initial concentration of NAD^+ was 100 µM,

i. What was the initial concentration of NADH?

Time = 0 min, Abs_{340nm} =0. Therefore, the initial concentration of NADH is (0.000)

ii. What was the final concentration of NADH after 5 min?

Time = 5 min, Abs_{340nm} =0.390. Therefore, the final concentration of NADH is = 0.390 ÷ 6620 M^{-1} = **5.891×10^{-5} M**

iii. How long did it take for half of the NAD^+ to be reduced to NADH?

Total amount of NAD^+ = 100 µM, half of it is **50 µM**

The absorption of a solution containing 50 µM NADH will be → Abs = [NADH] × 6620 M^{-1}, Abs = $50\times10^{-6} \times 6620\ M^{-1}$ = **0.331**

Therefore, it took **3 min** for half of the NAD+ to be reduced to NADH.

1 + 1 + 2 = 4 marks

c. What would happen if instead of following the reaction at 340 nm the colorimeter was set up to measure absorbance at 260 nm? Could you follow the conversion of NAD+ into NADH? Explain.
If the colorimeter was set at 260 nm it would not be possible to follow the conversion of NAD+ into NADH given that both species show a strong absorption at this wavelength [1 mark]. Therefore, the conversion of one into the other would not change much the reading of the spectrophotometer/absorption measured [1 mark].

2 marks

END OF QUESTION AND ANSWER BOOK